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ORIGINAL RESEARCH

Integrating deep learning and traditional image enhancement techniques for underwater image enhancement

Zhenghao Shi | **Yongli Wang** | **Zhaorun Zhou** | **Wenqi Ren**

School of Computer Science and Engineering, Xi'an University of Technology, Xi'an, China

Correspondence
Z. Shi, School of Computer Science and Engineering, Xi'an University of Technology, No. 5, Jinhua South Road, Xi'an 710048, Shaanxi, China.
Email: ylshi@xaut.edu.cn

Abstract

Underwater images usually suffer from colour distortion, blur, and low contrast, which hinder the subsequent processing of underwater information. To address these problems, this paper proposes a novel approach for single underwater images enhancement by integrating data-driven deep learning and hand-crafted image enhancement techniques. First, a statistical analysis is made on the average deviation of each channel of input underwater images to that of its corresponding ground truths, and it is found that both the red channel and the green channel of an underwater image contribute to its colour distortion. Concretely, the red channel of an underwater image is usually seriously attenuated, and the green channel is usually over strengthened. Motivated by such an observation, an attention mechanism guided residual module for underwater image colour correction is proposed, where the colour of the red channel of the underwater image and that of the green channel is compensated in a different way, respectively. Coupled with an attention mechanism, the residual module can adaptively extract and integrate the most discriminative features for colour correction. For scene contrast enhancement and scene deblurring, the traditional image enhancement techniques such as CLAHE (contrast limited adaptive histogram equalization) and Gamma correction are coupled with a multi-scale convolutional neural network (MSCNN), where CLAHE and Gamma correction are used as complement to deal with the complex and changeable underwater imaging environment. Experiments on synthetic and real underwater images demonstrate that the proposed method performs favourably against the state-of-the-art underwater image enhancement methods.

1 | INTRODUCTION

The visual quality of underwater images is of great importance for underwater scene understanding and recognition. However, due to light selective absorption and scattering in water, underwater images usually suffer from serious colour distortion, blurring, low contrast and brightness, which severely hinder the performance of underwater scene understanding and recognition.

Various methods have been developed in past years to improve the visual quality of underwater images [1–3]. These methods can be roughly grouped as non-model-based, model-based, and deep learning-based methods. The non-model-based methods such as histogram equalization, homomorphic filtering, and retinal-based enhancement algorithms, do not consider the image degradation mechanism, and directly process the pixels by highlighting the features of interest in the image and suppressing some unimportant features [2]. However, without considering the image degradation mechanism, colour distortion and supersaturation may remain occur, resulting in the loss of some image information. Different from non-model-based methods, physical model-based methods consider the imaging optical physical model, such as underwater image enhancement method based on wavelength compensation, blue-green channel, and red channel [3]. Different from traditional image enhancement and restoration, deep learning-based methods explore constraints from massive training data to restore clear images. Though significant progress achieved, deep learning based underwater image enhancement is still challenging due to the complexity of degradation mechanism for underwater imaging which contains local details lost and global colour distortion over all image pixels. In addition, existing deep learning-based

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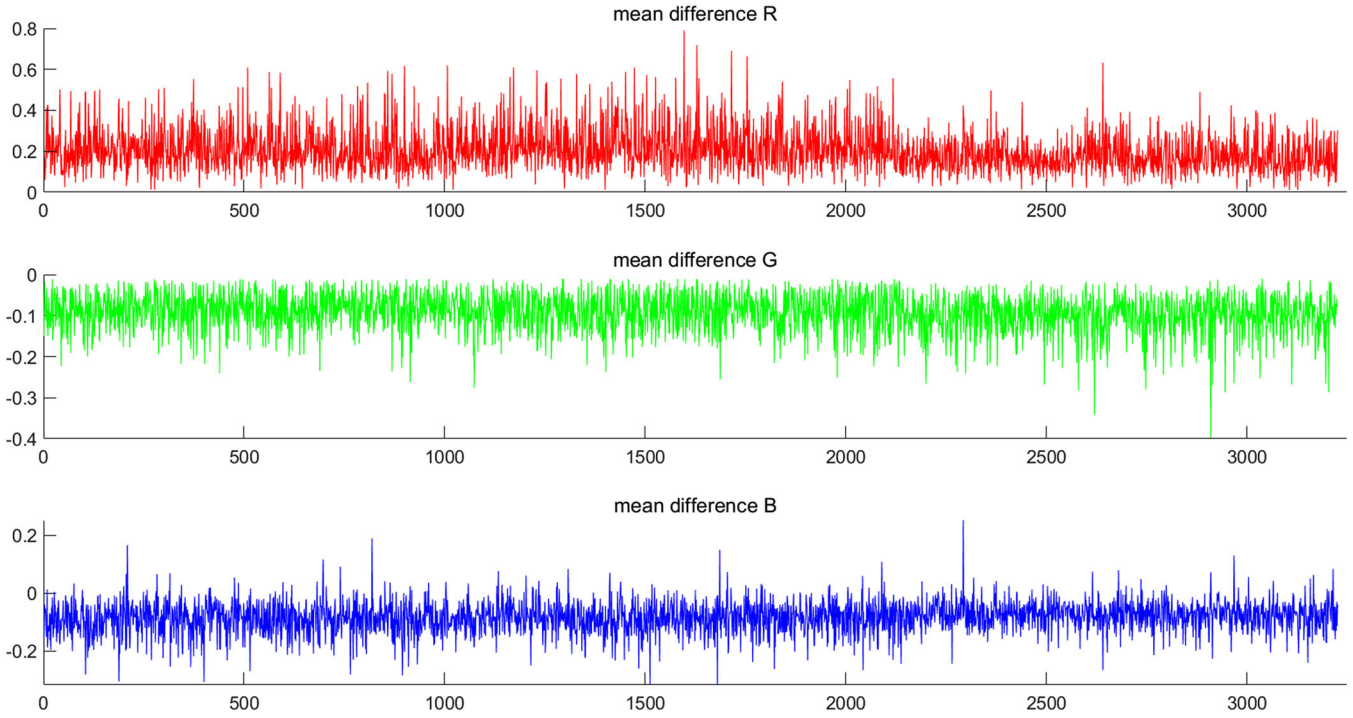


FIGURE 1 A statistical analysis of the average deviation of each channel of input underwater images to that of the ground truth

methods require large numbers of parameters to learn the complex mapping functions. This limits the potential value of these deep learning methods in practical applications.

As we all know that different wavelengths of light are attenuated at different rates in water. Concretely, the red colour light travels the shortest due to its longest wavelength, and the blue colour travels the longest in the water due to its shortest wavelength. This is the reason that the colour of underwater image is deviated to its inherent colour. Figure 1 shows a statistical analysis of the average deviation of each channel of input underwater images to that of the ground truth. As it can be seen that the average deviation of the blue channel of underwater images to that of its corresponding ground truth is relatively narrow, which implies that the blue colour of the underwater image is well preserved. However, the value of the average deviations of the red channel of underwater images to that of its corresponding ground truth are all above 0, which implies that the red channel information of underwater images is seriously attenuated. While the value of the average deviations of the green channel are all under 0, which implies that the green channel information of underwater image is over strengthened. Based on such an observation, to well correct the colour of the input underwater image, we propose to perform colour compensation both for the red channel and the green channel of underwater images which is different from existing methods where colour compensation is only done in the red channel of underwater images. For scene contrast enhancement and scene deblurring, we employ a multi-scale convolutional neural network (MSCNN) in this work. Of which we use three 3×3 convolution kernels to extract deeper features while reducing a certain amount of parameters in the network, and use two 5×5 mid-scale convolution kernels to extract more essential features of the image, and using a 7×7 convolution kernel to avoid excessive parameters. Finally, the obtained information weights of different degrees are combined through a combination of multiplication and addition to achieve the purpose of enhancing the overall details of the image, CLAHE and Gamma correction are also coupled to adjust the histogram distributions and illumination of the image.

The main contributions of this article are as follows:

1) We propose such a prior that both the red channel and the green channel of an underwater image contribute to its colour distortion. Concretely, the red channel of an underwater image is usually seriously attenuated, and the green channel is usually over strengthened.
2) We propose an attention mechanism guided residual module for underwater image colour correction. Where the colour of the red channel of the underwater image and that of the green channel is compensated in a different way, respectively. By coupled with an attention mechanism, the residual network modular can adaptively extract and integrate the most discriminative features for colour correction.
3) We coupled the traditional image enhancement techniques with a multi-scale convolutional neural network (MSCNN) for scene contrast enhancement and scene deblurring. It explores the complementary merits between domain knowledge of traditional image enhancement techniques and deep neural network; therefore, it could deal with the complex and changeable underwater imaging environment.

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4) Our proposed method achieves state-of-the-art performance on several benchmarks in terms of both visual quality and quantitative metrics.

The rest of the paper is organized as follows. Section 2 presents the existing deep learning-based underwater image enhancement algorithms and insights into the network. Section 3 describes our underwater image enhancement method and the introduction of the network. Section 4 gives the experimental quantitative and qualitative results and analysis. Section 5 concludes this paper.

2 | RELATED THEORIES

According to whether it depends on the specific imaging model and whether it adopts deep learning technology, underwater image enhancement methods can be divided into non-physical model methods, physical model-based methods and deep learning methods.

To improve the visual quality of underwater images, instead of relying on the specific imaging model, the non-physical model methods directly adjusting the pixel value of the image. For example, in literature [4, 5, 6], instead of considering the causes of colour distortion, they assume that the average value of all image pixels tends to the same gray value based on the gray world hypothesis. With such an assumption, the influence of ambient light is removed from the input image by white balance method to obtain the enhanced image. In literature [7], by simulating the human visual system, the retinal method based on colour constancy theory is used to enhance the underwater image according to the characteristics of each image channel. Though this kind of method can improve the visual quality of underwater image to a certain extent, it is easy to introduce colour deviation and artefacts because of without considering the optical characteristics of underwater image imaging.

Different from non-physical model methods, physical model-based methods treat the underwater image enhancement as an ill-posed inverse problem, and explore the physical imaging model and various prior to estimate the desired results from degraded input images. For example, Shin et al. [8] used the convolution networks to estimate ambient light and realized underwater image colour correction based on physical model. Codruta et al. [9] estimated the light attenuation according to the red channel, and then locally adjusted the pixel values of other colour channels by using the colour transfer function, so as to enhance the underwater image. Jamadandi et al. [10] corrected the distorted image colour using the wavelength correction depth convolution network parameters. The method based on physical model depends on the mathematical modeling of underwater imaging process, and the model is usually based on certain assumptions a priori which has certain limitations, and the complexity of parameter estimation algorithm is also very high.

Compared with various traditional underwater image enhancement methods mentioned above, deep learning-based methods can directly learn the mapping relationship between underwater image and transmission image or clear image from a large number of training samples, and achieves better enhancement results than that with traditional methods [1]. According to the neural network models, deep learning-based methods can be divided into two categories: underwater image enhancement method based on convolution neural networks and underwater image enhancement method based on generative adversarial networks.

The underwater image enhancement method based on convolution neural networks can also be divided into two categories. One is to use the network to directly learn the mapping relationship between degraded image and clear image to enhance the underwater image. For example, Wu et al. [11] proposed a novel two-stage underwater image convolutional neural network based on structure decomposition for underwater image enhancement, contained high-frequency and low-frequency networks. For example, Li et al. [12] employs the underwater imaging physical model to synthesize the underwater image degradation data sets covering different water types and degradation degrees firstly. Then the synthesized underwater images are used to train a convolution networks model to reconstruct clear underwater images. Perez et al. [13] proposed a single underwater image enhancement method based on the imaging optical model and the convolutional neural networks. This method can take a single image as input to generate an image with good image restoration quality. Hu et al. [14] proposed an underwater image enhancement method based on a double convolution neural networks architecture. The method is composed of transmittance estimation network and global ambient light estimation network. It learns the relationship between underwater scene and its corresponding blue channel transmission diagram and global ambient light, respectively, and then restores and enhances underwater image by using underwater optical imaging model. Hou et al. [15] proposed an underwater image enhancement method based on joint residual learning. The underwater image enhancement task is modeled as learning transmission map and scene residual at the same time. The model includes data-driven residual structure for transmission map estimation and knowledge driven scene residual calculation method for underwater lighting balance. Therefore, this method can integrate the prior knowledge and data information to restore the underwater image. Huang et al. [16] combined colour correction method with deblurring network to improve the quality of underwater image. This kind of methods can reduce the learning difficulty of the network and is easy to obtain better restoration results. However, they need paired underwater degraded image and corresponding clear image as training data.

Underwater image enhancement based on generative adversarial networks is also a research hotspot in recent years. For example, by simulating underwater imaging degradation (absorption and scattering), C et al. [17] proposed a novel method for generating synthetic underwater images considering revised image formation model and use the generated synthetic underwater images to train a CGAN towards restoration of degraded underwater images. Yang et al. [18] also develop a generative adversarial network- based method for underwater

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image enhancement. Though it has good perception ability for sample distribution, it still needs paired labeled images as training samples. Chen et al. [19] proposed an underwater image restoration method based on encoder decoder structure. Specifically, the generator adopts an encoder-decoder structure to generate a composite image at the original resolution through 9 residual blocks and deconvolution. In order to maintain the image content and remove underwater noise at the same time, a multi branch discriminator including anti-branch and critical branch is adopted. In addition to adversarial learning, a new dark channel priority loss is also used for constraint generator to produce realistic visual effects. This method can adaptively improve the underwater visual quality in real time to some extent, and produce overall superior restoration performance. However, due to the small number of layers of its generation model, it is difficult to obtain the contextual information of different images in a variety of conditional information, so the generated images are easy to lose content and structure information. Aiming at the problem that the existing deep learning methods cannot well solve the poor restoration caused by absorption and scattering because they do not fully consider the underwater physical deformation process, literature [20] proposed a two-stage underwater image restoration network guided by attenuation coefficients. Firstly, the underwater imaging physical model is embedded into the deep learning network to deal with the horizontal distortion, and the attenuation coefficients are used to characterize the water type information to guide the parameter estimation of the underwater imaging physical model. Then, based on a new attenuation coefficient priority attention block, the RGB channel feature map of the vertical distortion image is adaptively corrected to reconstruct a clear underwater image. Hong et al. [39] proposed a novel weakly supervised underwater image enhancement method, a novel generative adversarial network (GAN)-based architecture to enhance underwater images by unpaired image-to-image transformation from domain X to domain Y.

3 | METHODOLOGY

Based on above analysis, we argue that utilizing only handcrafted or deep learning method cannot well handle the tough underwater environment. Therefore, we choose to take advantages of both kinds of methods for underwater images enhancement. Figure 2 illustrates the framework of the proposed method. It adopts a modular structure and begins from the decomposition of the input image. The input image is first decomposed as its corresponding R, G, B channel image, and then colour compensation is performed on the R channel and the G channel of the input underwater image utilizing attention guided residual module, respectively, which is different from existing methods that do colour compensation only for the red channel of underwater images. This strategy is motivated such a prior that both the red channel and the green channel of an underwater image contribute to its colour distortion. Concretely, the red channel of an underwater image is usually seriously attenuated, and the green channel is usually over strengthened, as analysed in Figure 1. The brightness of the B channel of the input image is also enhanced with a residual convolution modular at the same time. By coupled with an attention mechanism, the residual network modular can adaptively extract and integrate the most discriminative features for colour correction. After colour correction, scene contrast enhancement and scene deblurring are followed. For scene contrast enhancement and scene deblurring, a multi-scale convolutional neural network (MSCNN) coupled with CLAHE and Gamma correction, is developed. By exploring the complementary merits between domain knowledge of traditional image enhancement techniques and deep neural network, it could deal with the complex and changeable underwater imaging environment.

In following subsections, we detail the key components of our method, including colour compensation using attention guided residual networks (Section 3.1), contrast enhancement using MSCNN coupled with CLAHE and Gamma correction (Section 3.2), and the loss function (Section 3.3).

3.1 | Colour compensation utilizing attention guided residual modules

From Figure 1, we can see that both the red channel and the green channel of an underwater image contribute to its colour distortion. Concretely, the red channel of an underwater image is usually seriously attenuated, and the green channel is usually over strengthened. Therefore, different from existing methods where colour compensation is only done in the red channel of underwater images, we propose to performed colour compensation both for the red channel and the green channel of underwater images. Concretely, the colour of the red channel of the underwater image and that of the green channel is compensated in a different way, respectively. Inspired by the Squeeze-and-Excitation method proposed by Hu et al. [21], we incorporate attention mechanism for colour correction.

Inspired by Ancuti C et al. [22], for red channel colour compensation, we develop the following network module:

$$Irc(x) = Ir(x) + CAB(Ir(x))\,SAB(1 - Ir(x)), \quad (1)$$

where x represents the pixel values of the input image in its red channel, I_r represents the red channel of the input image I, I_{rc} represents the enhanced red channel image, CAB stands for channel attention module and SAB stands for spatial attention module.

Figure 3a shows the structure of the attention guided residual module for red channel colour compensation. It consists of five branches, the first top branch is a long skip connection. The purpose of this branch is to preserve the data fidelity and address gradient vanishing during the colour correction processing. The second top branch is a $1 \times 1 \times C$ channel attention map generator that consists of a global average pooling layer, a linear transformation layer, and a Sigmoid layer. The third top branch is the input image, and it is multiplied with the output of the second top branch to obtain the channel feature map. Its purpose is to explicitly exploit the most discriminative fea-

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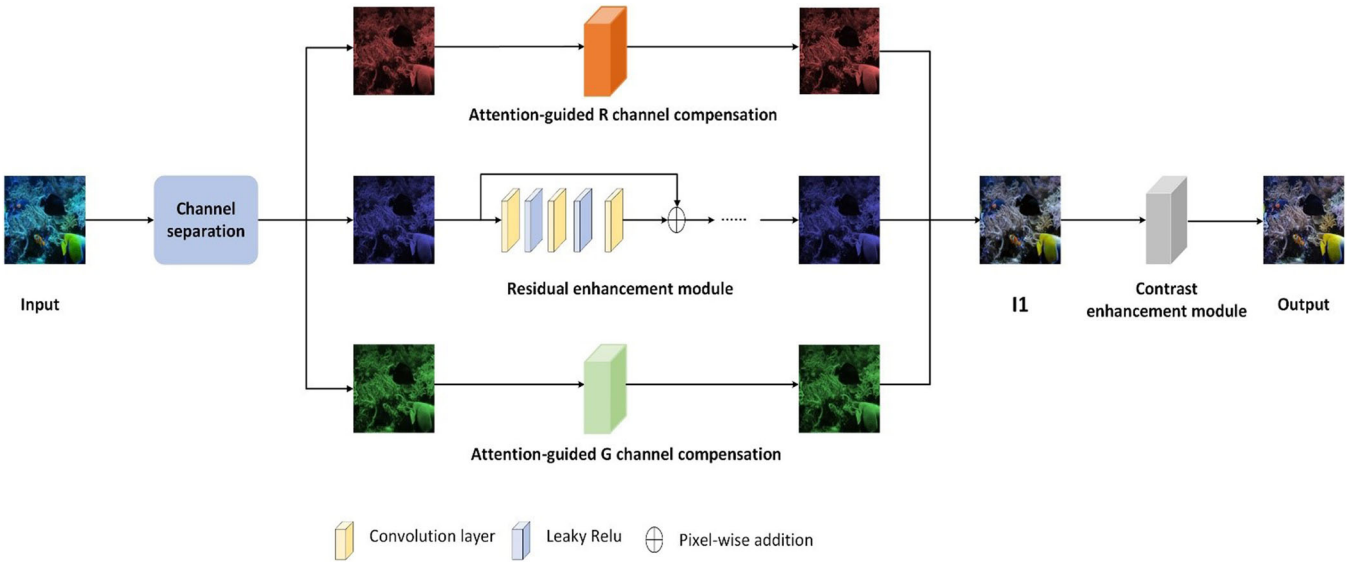


FIGURE 2 The framework of the proposed underwater image enhancement method. It adopts a modular structure and begins from the decomposition of the input image. The input image is firstly decomposed as its corresponding R, G, B channel image, and then colour compensation is performed on the R channel and the G channel of the input underwater image utilizing attention guided residual networks, respectively. After colour correction, contrast enhancement and scene deblurring are followed

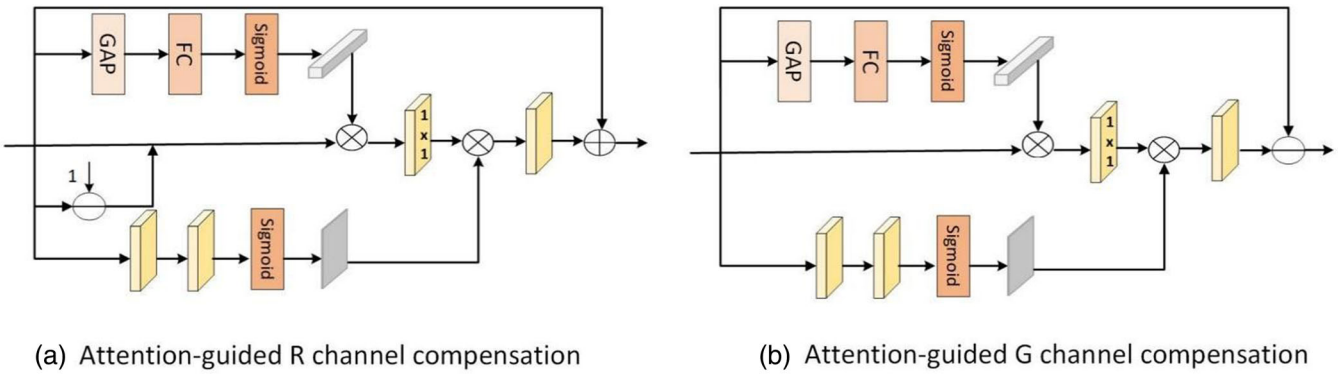


FIGURE 3 Attention guided colour correction network for the red channel and the green channel of underwater images. GAP stands for Global Average Pooling, and FC stands for a fully connected layer. (a) shows the red channel compensation network, and (b) shows the green channel compensation network

tures from red channel. To avoid over-assignment in areas with larger weights, the constrain of $1 - I_r(x)$ [22] is introduced to above operation by the second bottom branch. The first bottom branch is a H × W × 1 spatial attention map generator that consists of two 3 × 3 convolutional layers and a Sigmoid layer. We utilize the spatial attention map as a feature selector to weight the importance of different spatial positions of the features. The high-quality degradation pixels (the pixels with larger spatial attention map values) are assigned higher weights. With the combination of produced spatial attention map with the channel feature map, a colour compensated red channel image could be obtained.

For green channel colour compensation, we develop the following network module:

$$Iga(x) = Ig(x) - CAB(Ig(x))\,SAB, \tag{2}$$

where x represents the pixel values of the input image in its green channel, I_g represents the green channel of the input image I, I_{ga} represents the enhanced green channel image, CAB stands for channel attention module, SAB stands for spatial attention module.

Figure 3b shows the structure of the attention guided residual module for G channel colour compensation. As can be observed, this network structure is very similar to that used for red channel colour compensation. The different between these two modules is that the constrain of $1 - I_r(x)$ has not be introduced to, and since the green information is relatively abundant, part of the green information is subtracted from the input to obtain the compensated green channel.

By coupled with an attention mechanism, the residual network modular can adaptively extract and integrate the most discriminative features for colour correction.

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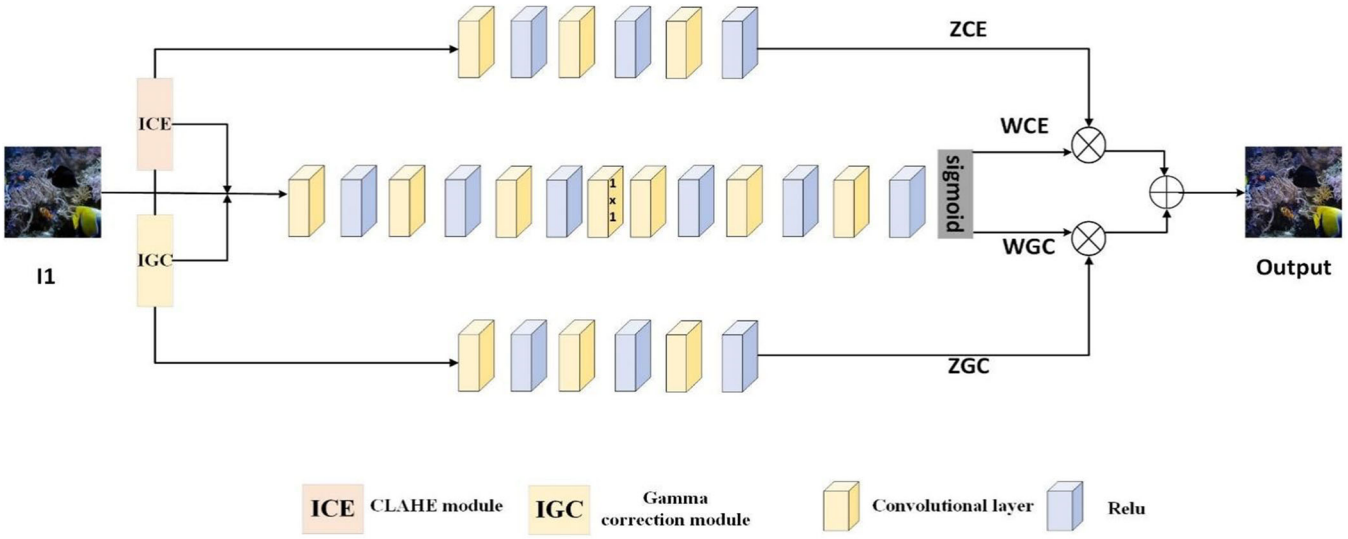


FIGURE 4 The contrast enhancement network for detail preservation. I_1 is the image after colour correction, I_{CE} is the image of I_1 after CLAHE algorithm and I_{GC} is the image of I_1 after Gamma correction algorithm. Z_{CE} and Z_{GC} are the weight values obtained by the first and third branches, W_{CE}, W_{GC} are the weight values predicted by the second branch.

3.2 | Contrast enhancement and scene deblurring using MSCNN combining with CLAHE and the gamma algorithm

For contrast enhancement and scene deblurring, a multi-scale convolutional neural network (MSCNN) is developed in this section. Figure 4 shows the structure of the proposed MSCNN. It consists of three branches. The top branch consists of a 3 × 3 convolutional layer, a 5 × 5convolutional layer, and a 7 × 7 convolutional layer. The purpose of the top branch is to eliminate the colour cast and artefacts introduced by the CLAHE algorithm. The middle branch consists of a 3 × 3 convolutional layer, a 5 × 5 convolutional layer, a 7 × 7 convolutional layer, a 1 × 1 convolutional layer, and a Sigmoid layer, its purpose is to make the output value in [0,1]. Then divide the weight value into two equally, one of which is weight value W_{CE}, and the other is weight value W_{GC}. The structure of the bottom branch is the same as that of the top one. The output of the top branch and that of the middle branch, as well as the output of the middle branch and that of the bottom branch, are fused by element multiplication, respectively. Finally, the above two fused outputs are combined to obtain the final enhanced image by an addition operation as formulated as follows:

$$I_{OUT} = Z_{CE} \times W_{CE} + Z_{GC} \times W_{GC}, \tag{3}$$

where I_{OUT} is the final output image, Z_{CE} and Z_{GC} are the weight values obtained by the first and third branches, and W_{CE}, W_{GC} are the weight values predicted by the second branch.

As we well know that the histogram distributions of natural-scene images are wider and more consistent while the histogram distribution of each colour channel of the underwater image is shifted in a horizontal direction. Inspired by this prior, we think to achieve better contrast and brightness, we can adjust the histogram distributions of underwater images. For this purpose, we coupled CLAHE and Gamma correction with aforementioned multi-scale convolutional neural network. Of which the gamma correction is adopted to balance the illumination and normalize the non-uniform illumination, and CLAHE is used for further enhance the contrast of scene. It explores the complementary merits between domain knowledge of traditional image enhancement techniques and deep neural network, therefore it could deal with the complex and changeable underwater imaging environment.

3.3 | Loss function

For the training of deep neural networks, the following three loss functions are widely used currently:

3.3.1 | L1 loss

L_1 loss is used to measure the minimization of the sum of absolute differences between the target value y_i and the corresponding estimated value $f(x_i)$, and defined as:

$$L_1 = \sum_{i=1}^{n} \left| y_i - f(x_i) \right|. \tag{4}$$

3.3.2 | Perceptual loss

Perceptual loss is defined as the distance between the feature representations of predicted RGB image and ground-truth image, and is used for expressing the information of colour and semantics. Due to the deep layer has a better understanding of semantic information and can fully preserve the image content and overall spatial structure, and the shallow layer is

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more sensitive to colour and texture, we select layer 4_3 from VGG-19 [23] to make it sensitive to both colour and semantics. The definition of our Perceptual loss is described as:

$$L_{per} = \sum_{x=1}^{X} \sum_{y=1}^{Y} \left| \varphi_i J\ (x,y) - \varphi_i \bar{J}\ \ (x,y) \right|, \tag{5}$$

where $\varphi_i J(x,y)$ represents the convolutional layer of layer j, J is the reconstruction result, and $\bar{J}$ is the ground truth value.

3.3.3 | SSIM loss

We use the SSIM loss [36] to impose the structure and texture similarity on the predicted image. In this paper, we use gray images, which convert from RGB images, to compute SSIM score, and for each pixel the SSIM value is computed within a 11×11 image patch around the pixel. The formula is as follows:

$$\mathrm{SSIM}\ (x) = \frac{\left(2\mu_x\ (x)\ \mu_y\ (x) + c1\right)\left(2\sigma_{xy}\ (x) + c2\right)}{\left(\mu_x^2\ (x) + \mu_y^2\ (x) + c1\right)\left(\sigma_x^2\ (x) + \sigma_y^2\ (x) + c2\right)}, \tag{6}$$

where $\mu_y(x)$ and $\sigma_y(x)$ correspond to the mean and standard deviation of the RGB patch from the predicted image, $\mu_x(x)$ and $\sigma_x(x)$ are the same corresponds to ground-truth image. And $\sigma_{xy}(x)$ is cross covariance. We set $c1 = 0.02$ and $c2 = 0.03$. The SSIM loss is expressed as:

$$L_{SSIM} = 1 - \frac{1}{N} \sum_{i=1}^{N} SSIM\ (x_i)\,. \tag{7}$$

To achieve a good balance between visual quality and quantitative scores, following previous works [32, 33], we use the linear combination of the of L1 loss, SSIM loss, and the perceptual loss L_{per}, and the final loss L for training our network is expressed as:

$$L = \lambda_1\ (L_1 + L_{SSIM}) + \lambda_2 L_{per}, \tag{8}$$

where λ_1 and λ_2 are the parameters used to adjust the weight.

4 | EXPERIMENT

4.1 | Implementation details

4.1.1 | Data set

Two datasets are employed in this study. The first one comes from [37], it is an underwater dataset synthesized by CycleGAN, and contains a total of 6128 image pairs with ground-truth (undistorted underwater images) and distorted underwater images. And each image has a corresponding reference image. From which we selected 4000 image pairs for training, and 100 image pairs from the remaining images for testing. The another one is the UIEB dataset [24], which contains 60 challenging real underwater images without corresponding reference images. To further verify the effectiveness of our mode, we grouped these images into five categories: greenish underwater images, bluish underwater images, yellow underwater images, shallow underwater images, and limited illumination underwater images.

4.1.2 | Training settings

For network training, the Adam optimizer with a learning rate of 0.0002 was used, the batch size was set as 8, and the epoch is set as 50. the size of all input images was set as 256×256. All experiments were performed on AMD Ryzen 53600 6-core Processor with 16GB RAM, and NVIDIA Geforce GTX 1650 GPU using Pytorch as the deep learning framework.

4.1.3 | Compared methods

We compared our method with nine methods, including one traditional method (red-channel [25]) and eight deep learning based method (BICE [26] and ATMG [27], UWCNN [28], GDCP [29], MLSD [30], MLFcGAN [31], BR [32] and MTGM [33].) For all compared methods, we used the released codes to produce their results.

4.1.4 | Evaluation metrics

For quantitative evaluation the performance of different methods on the synthetic datasets, the full-reference metrics of PSNR (Peak Signal to Noise Ratio, PSNR) [40] and SSIM (Structural Similarity, SSIM) [41] are employed. Both of the quantitative measures are calculated using the luminance channel. A higher SSIM indicates greater image structure similarity between a result image and its ground truth (For the ground truth, the SSIM equals 1). A high value of PSNR indicates a better enhancement performance. Performance of different methods on real-world images is evaluated visually since the ground truth images are not available.

For quantitative evaluation the performance of different methods on real underwater images, since no ground truth images are available, we adopt four widely used reference-free metrics, that is, UIQM (Underwater Image Quality Measure [34] and UCIQE (Underwater Colour Image Quality Evaluation) [35] for evaluating real-world underwater image quality. PCQI (patch-based contrast quality index) [38] to evaluate the contrast of the enhanced underwater images. NIQE (Natural Image Quality Evaluator) the evaluate quality of the images. A lower NIQE score represents a better human visual perception. A higher PCQI score indicates the enhanced image has better contrast, while a larger UIQM shows better colour and contrast, and is more suitable for human vision. A larger UCIQE value demonstrates the enhanced result has well balance among chroma, luminance, and saturation. The UIQM comprises

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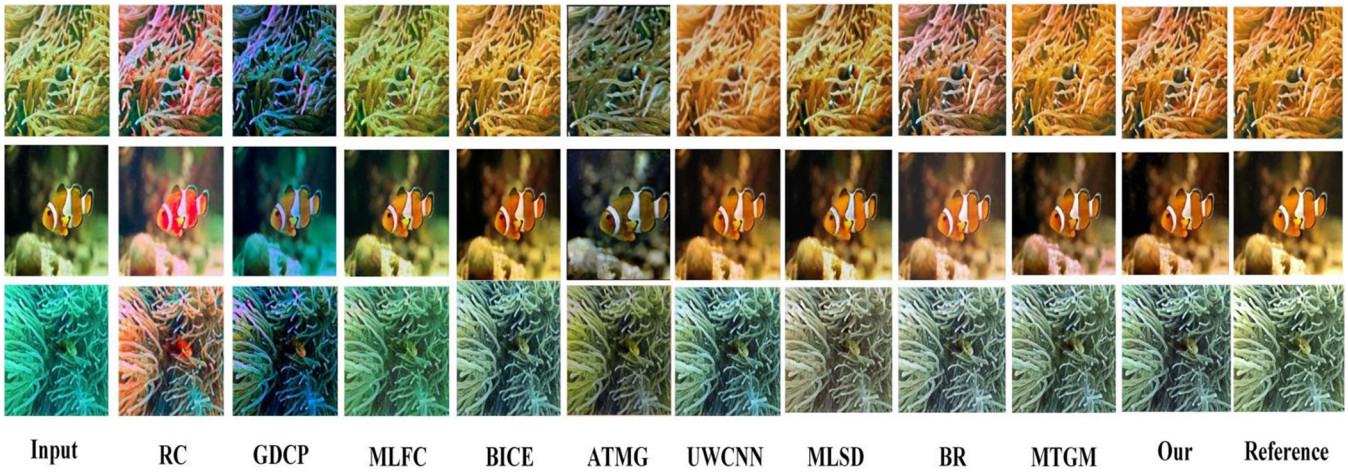


FIGURE 5 The comparison of the visual results of all methods on the synthetic dataset with reference. From left to right are raw images, and the results of red-channel [25], GDCP [29], MLFcGAN [31], BICE [26], ATMG [27], UWCNN [28], MLSD [30], BR [32], MTGM [33], Our method and ground Reference

three properties of underwater images, such as the underwater image colourfulness measure (UICM), the underwater image sharpness measure (UISM), and the underwater image contrast measure (UIConM). The UIQM can be written as:

$$UIQM = c1 \times UICM + c2 \times UISM + c3 \times UIConM. \quad (9)$$

Among them, we set the scale factors C1 as 0.0282, C2 as 0.2953 and C3 as 3.5735, which are consistent with the values in the original paper. The higher the UIQM value, the better the colour balance, sharpness, and contrast of the underwater image.

The UCIQE utilizes a linear combination of chroma, saturation and contrast to quantify the nonuniform colour cast, blurring, and low contrast. Its formula is as follows:

$$UCIQE = c1 \times \sigma c + c2 \times conl + c3 \times \mu 3, \quad (10)$$

where σc represents the standard deviation of chroma, conl represents contrast, and $\mu 3$ represents saturation. c1, c2, and c3 represent the three weight coefficients. We set c1 as 0.4680, c2 as 0.2745, and c3 as 0.2576, which are consistent with the parameter Settings in the original paper. A higher UCIQE value indicates the image has better balance among the chroma, saturation and contrast.

4.2 | Experiment results

4.2.1 | Experiment on datasets with reference

Figure 5 shows part comparing results on synthetic datasets. As it can be observed that the red-channel [25] effectively improves the contrast but the greenish has not been well removed. And GDCP [29] cannot work on the bluish underwater. Results with MLFcGAN [31] have a colour cast. Compared with above mentioned traditional methods, the latest proposed methods, including BICE [26], ATMG [27], UWCNN [28], MLSD [30], BR [32] and MTGM [33], all could obtain better enhancement effect. However, we noticed that all the methods fail to recover the complete scene structure. By contrast, our results are the closest one to the ground-truth, where the haze and colour casts (especially the background colour cast) of the underwater images are well removed. In addition, our results even achieve better visual quality than the corresponding reference images (e.g., less noise and better details).

TABLE 1 Quantitative analysis results on synthetic dataset

Method	PSNR(dB)	SSIM
Input	16.85	0.692
RC [25]	16.91	0.701
GDCP [29]	19.02	0.773
MLFC [31]	22.27	0.798
BICE [26]	22.36	0.815
ATMG [27]	23.15	0.821
UWCNN [28]	21.11	0.806
MLSD [30]	23.27	0.833
BR [32]	23.06	0.829
MTGM [33]	23.50	0.840
Ours	**25.67**	**0.891**

Table 1 shows quantitative analysis results on synthetic data, among all compared methods, our method obtained the highest score both for PSNR (25.67)and SSIM (0.891).

4.2.2 | Experiment on real datasets

Figure 6 shows part comparison on limited illumination underwater images. It can be seen that, in the input images, the structural details of the underwater scene are significantly hides

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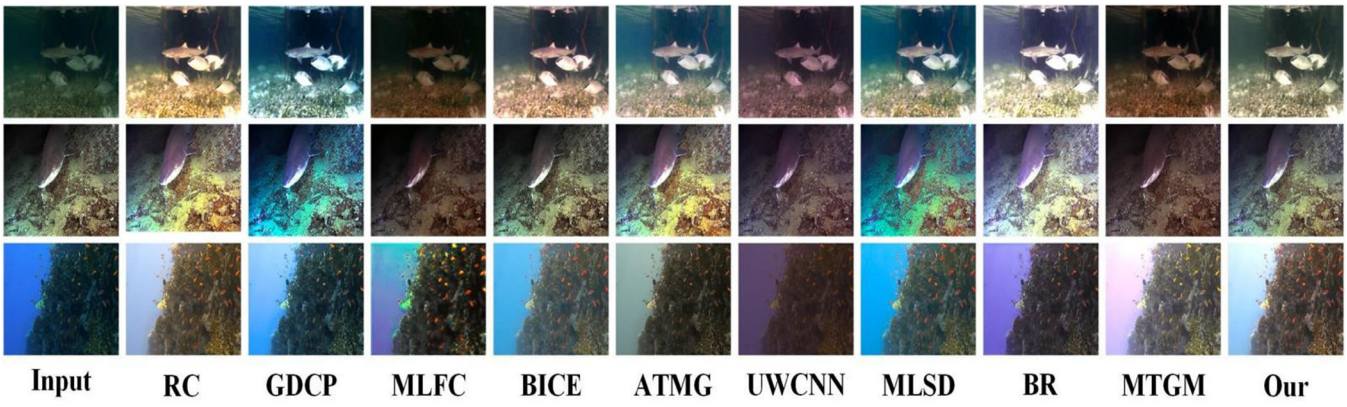


FIGURE 6 The comparison of the visual results of all methods on limited illumination underwater images. From left to right are raw images, and the results of red-channel [25], GDCP [29], MLFcGAN [31], BICE [26], ATMG [27], UWCNN [28], MLSD [30], BR [32], MTGM [33], Our method

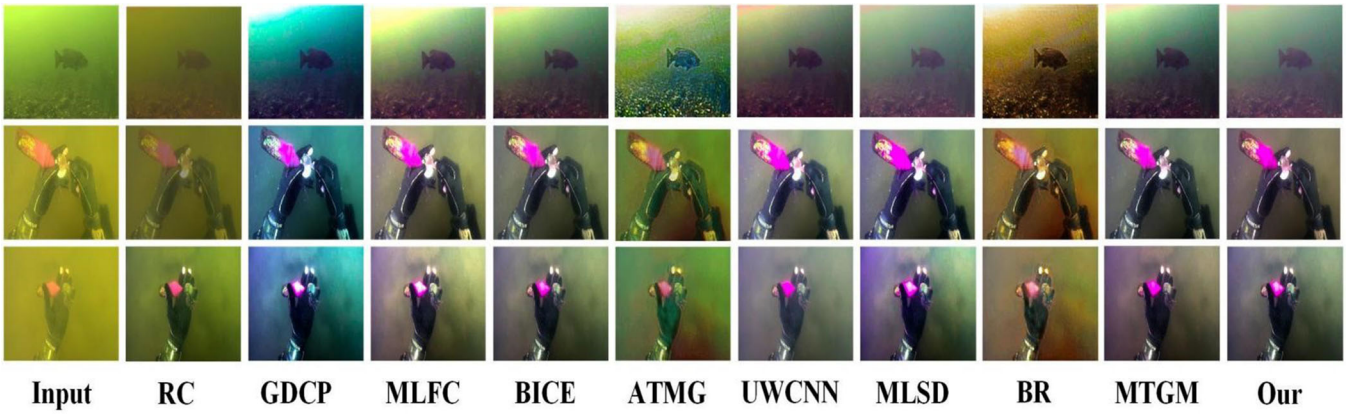


FIGURE 7 The comparison of the visual results of all methods on yellow underwater images. From left to right are raw images, and the results of red-channel [25], GDCP [29], MLFcGAN [31], BICE [26], ATMG [27], UWCNN [28], MLSD [30], BR [32], MTGM [33], Our method

by the limited illumination. In terms of colour, GDCP [29], MLFC [31], BICE [26], UWCNN [28], MLSD [30], BR [32], and MTGM [33] introduce extra colour artefacts. Except our method, all other compared method under-enhanced the image. In comparison, our method effectively improves the contrast without obvious over enhancement. This demonstrates the advantage of our method for limited illumination underwater images.

Figure 7 shows part comparison on yellow underwater images. As it can be observed that all competing methods cannot achieve satisfactory results under such a condition, and GDCP [29], MLFC [31], BICE [26], UWCNN [28], MLSD [30], BR [32], and MTGM [33] even introduce artificial colours. In contrast, our method obtained the most natural colour with enhanced details.

Figure 8 shows part comparison on bluish underwater images. As presented, RC [25], ATMG [27], and BR [32] produce obvious colour deviations. while GDCP [29], MLFC [31], and BICE [26] under-enhanced the images. In comparison, our method not only recovers the relatively realistic colour but also enhances details.

Figure 9 shows part comparison on greenish underwater images. RC [25], GDCP [29], MLFC [31], UWCNN [28] produce colour casts and artefacts, and over-enhanced. Compared with the rest of the methods, our method can remove the green tone better and does not produce artefacts, and achieve better results visually.

Figure 10 shows part comparison on shallow underwater images. GDCP [29], MLFC [31], UWCNN [28], ATMG [27] produce artefacts and over-enhanced. Compared with other methods, our method can well solve the problem of chromatic aberration of underwater images under illumination.

Figure 11 shows the statistics analysis of the mean difference between the enhanced underwater image and the corresponding real value. It can be seen that the mean difference of the R, G, and B channels are all around 0. This indicates that the enhanced result is basically consistent with the real value.

Table 2 shows the average values of UCIQE, UIQM, PCQI [38], and NIQE, which are obtained by evaluating Figures 6–10. As it can be seen that of all compared methods, our proposed method has obtained the highest average value for all metrics than that with other methods. The proposed method has better

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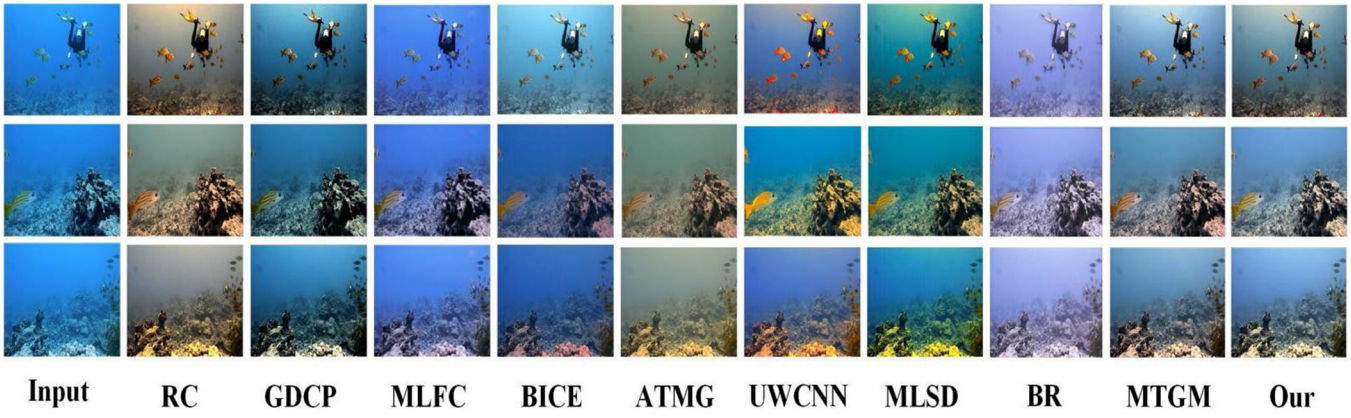


FIGURE 8 The comparison of the visual results of all methods on bluish underwater images. From left to right are raw images, and the results of red-channel [25], GDCP [29], MLFcGAN [31], BICE [26], ATMG [27], UWCNN [28], MLSD [30], BR [32], MTGM [33], Our method

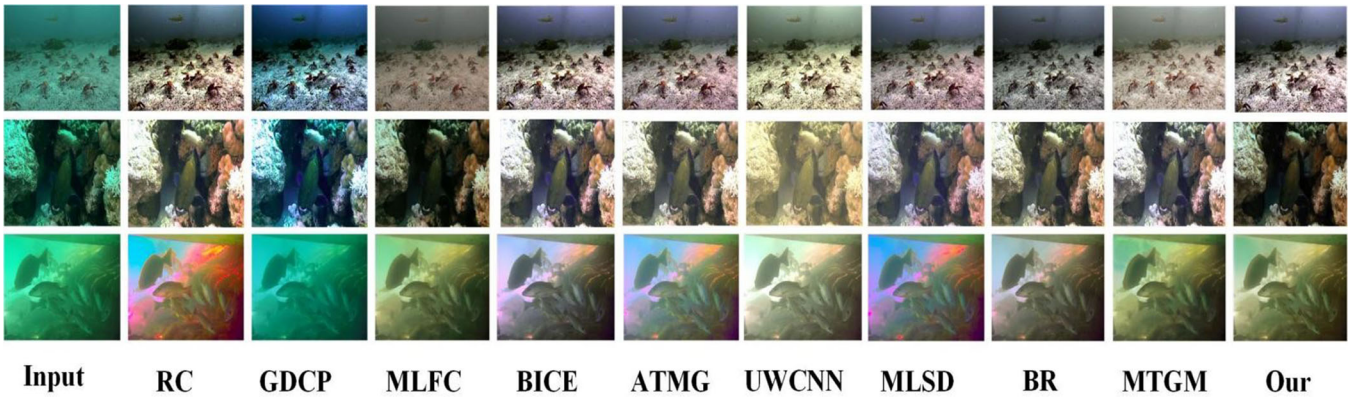


FIGURE 9 The comparison of the visual results of all methods on greenish underwater images. From left to right are raw images, and the results of red-channel [25], GDCP [29], MLFcGAN [31], BICE [26], ATMG [27], UWCNN [28], MLSD [30], BR [32], MTGM [33], Our method

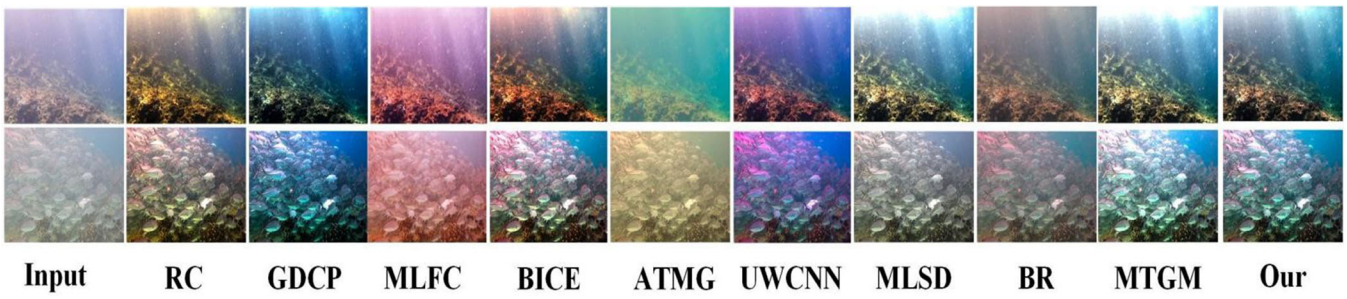


FIGURE 10 The comparison of the visual results of all methods on shallow underwater images. From left to right are raw images, and the results of red-channel [25], GDCP [29], MLFcGAN [31], BICE [26], ATMG [27], UWCNN [28], MLSD [30], BR [32], MTGM [33], Our method

subjective evaluation results in respect to colour restoration, edge-preserving, artefacts-free results, no halo effects, and texture smoothing, as can be observed from the corresponding figures.

Table 3 shows the average values of PSNR and SSIM obtained by evaluating Figures 6–10. From Table 3, it can be seen that of all compared methods, the proposed method has obtained the highest average value of PSNR and that of SSIM. Further, the average PSNR values obtained by the following methods - BR [32] and MLSD [30] has a higher average PSNR value and a higher SSIM value, as shown in Table 3. Even though the methods BR [32] and MLSD [30] have higher PSNR value, the subjective evaluation results are unsatisfactory. The average PSNR value of red-channel [25] ranked the last. The results obtained by red-channel [25] are with poor colour restoration. The average PSNR value of GDCP [24] ranked

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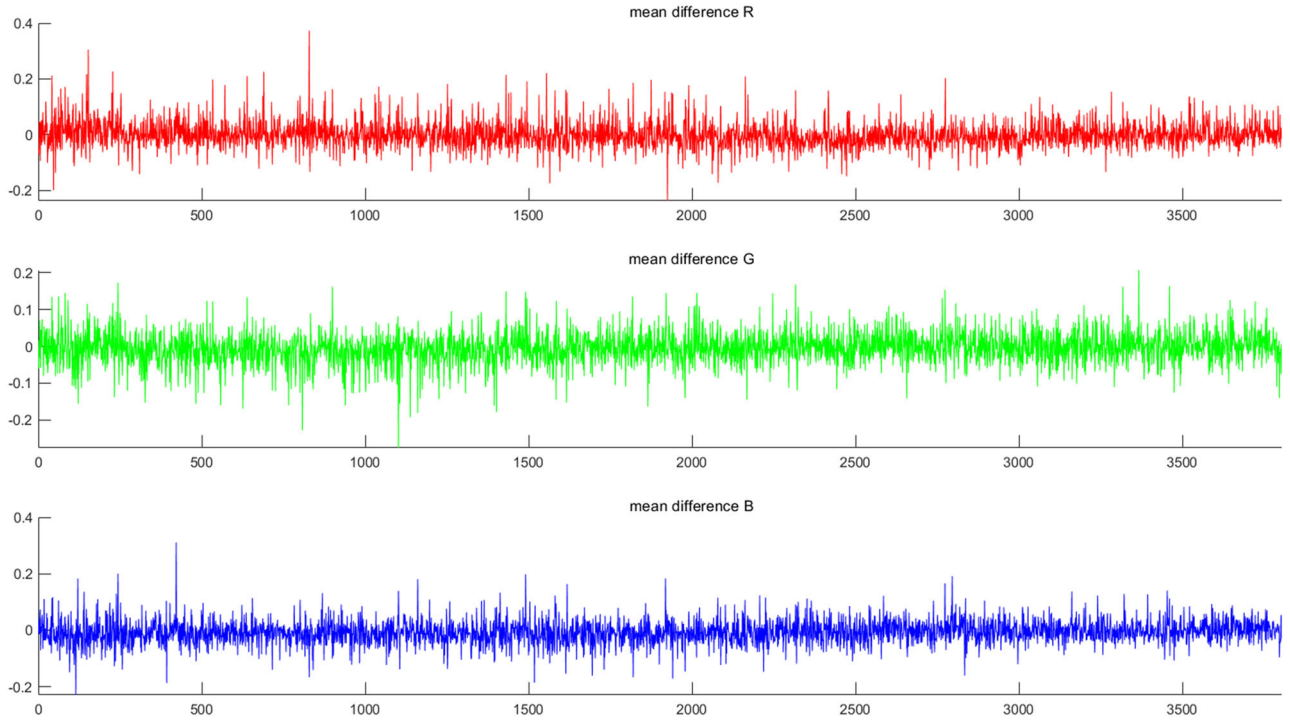


FIGURE 11 A statistical analysis of the average deviation of each channel of enhanced underwater images to that of the ground truth

TABLE 2 Average quantitative analysis results on real dataset

Method	UCIQE	UIQM	PCQI	NIQE
RC [25]	0.435	1.103	0.672	5.787
GDCP [29]	0.518	1.257	0.831	5.441
MLFC [31]	0.534	1.260	0.963	5.360
BICE [26]	0.577	1.465	1.103	4.982
ATMG [27]	0.582	1.409	1.141	5.145
UWCNN [28]	0.554	1.515	0.987	4.914
MLSD [30]	0.583	1.504	1.094	4.683
BR [32]	0.572	1.429	1.077	4.707
MTGM [33]	0.611	1.517	1.115	4.371
Ours	**0.647**	**1.650**	**1.248**	**4.200**

TABLE 3 Average quantitative analysis results on real dataset

Method	PSNR(dB)	SSIM
RC [25]	15.37	0.584
GDCP [29]	15.81	0.587
MLFC [31]	19.83	0.771
BICE [26]	21.34	0.807
ATMG [27]	21.78	0.819
UWCNN [28]	20.00	0.801
MLSD [30]	22.49	0.793
BR [32]	23.57	0.814
MTGM [33]	23.72	0.826
Ours	**24.89**	**0.837**

nineth, the subjective evaluation results include white texture and lower colour restoration. The white texture leads to poor visibility in understanding the image details.

4.3 | Ablation study

To analyse the effect of each modular of our method, extensive ablation studies with several designed variants were performed in this section, the ablation study involves using or not using the following components:

- w/ Ce: the network has Ce modules on the basis of Part-I;
- w/ Rnet+ Gnet: both Rnet and Gnet are employed in our method;
- w/o Gnet: no Gnet used;
- w/o Rnet: no Rnet employed;
- w/o Rnet+ Gnet: both Rnet and Gnet are not employed in our method;

where 'Rnet' indexes the red channel compensation network module, 'Gnet' indexes the green channel compensation network module, and 'Ce' is the contrast enhancement module.

Figure 12 and Table 4 show the ablation results, respectively. From Figure 12, it can be seen that the proposed method 'w/ Ce' retains more detail with well recovered colour which is the closest one to the reference image. Further, without 'Ce'it is challenging to attain human visual perception, therefore 'Ce' is necessary for colour restoration. The proposed method 'w/o Rnet+ Gnet' failed to remove colour cast from the input image, whereas the method of 'w/ Rnet+ Gnet' has well removed colour cast from the input underwater images. In the proposed

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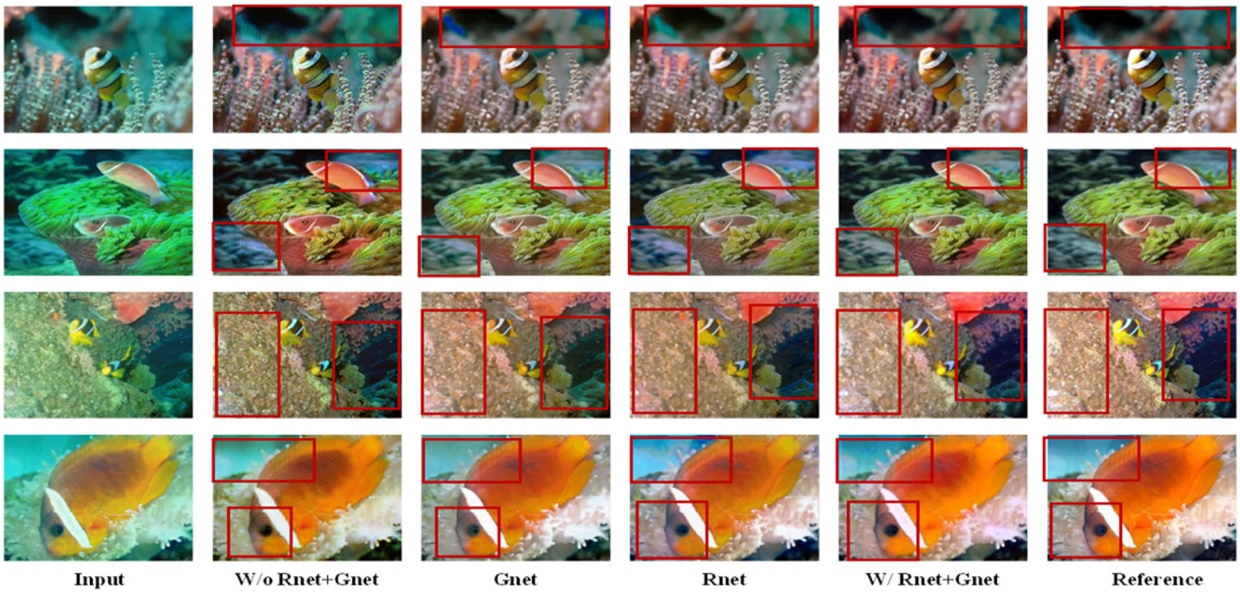


FIGURE 12 The comparison of the visual results of each module in the colour correction section. We have marked the parts where the difference in colour and detail is more obvious

TABLE 4 Ablation study on synthetic dataset

Combination module	PSNR(dB)	SSIM
w/o Gnet	21.36	0.8331
w/o Rnet	19.07	0.7905
w/o Gnet+Rnet	17.73	0.6902
w/ Gnet+Rnet	24.52	0.8592
w/ Ce	**25.67**	**0.8910**

TABLE 5 The experiments for the suitable parameter λ_1 and λ_2

λ_1	λ_2	PSNR	SSIM
0.1	0.1	21.16	0.814
0.1	1	20.02	0.726
0.1	10	19.87	0.701
1	**0.1**	**25.67**	**0.891**
1	1	22.75	0.807
1	10	20.14	0.743
10	0.1	19.36	0.778
10	1	19.15	0.740
10	10	19.01	0.719

method of 'w/o Gnet' or 'w/o Rnet', the image colour cast is insufficient to remove. The integration of 'Rnet' and 'Gnet' achieves more well image colour enhancement. The proposed method results in better colour restoration, artefacts free, edge preserving.

Table 4 shows the quantitative evaluation for various configurations with PSNR and SSIM based on Figure 12. From Table 4, it can be seen of all various configurations the proposed method shows the highest values in terms of PSNR and SSIM. Its PSNR is about 3dB higher than that of only one of the channels, indicating that the compensation module we proposed is effective.

We also analyse the setting of the hyperparameter λ_1 and λ_2 of the loss function (defined as Equation 4). The results are shown in Table 5. It can be seen from Table 5 that when λ_1 is set to 1 and λ_2 is set to 0.1, the evaluation index of the model is the best, so we set λ_1 to 1 and λ_2 to 0.1 in this paper.

5 | CONCLUSION

In this article, we propose a novel colour correct method by performing colour compensation both for the red channel and the green channel of underwater images. From the two perspectives of red channel enhancement and green channel attenuation, channel and spatial attention modules are used to complete colour correction. At the same time, the contrast enhancement network we proposed improves the overall contrast of the image and the brightness of the dark area. We conduct comparative experiments and ablation experiments on synthetic data sets and real underwater data sets, which proves that our method is better both in terms of indicators and vision.

And we conducted an experiment on the mean difference between the enhanced underwater image and its corresponding natural image. It can be seen from the figure that the mean difference of the three channels R, G, and B are all around 0, which is different from the degraded water at the beginning of the article. The comparison of the mean difference between the lower image and its corresponding natural image fully demonstrates the effectiveness of our method.

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CONFLICT OF INTEREST

We declare that no financial and personal relationships or organizations inappropriately affect our work and that no professional or other personal interests of any nature in any product, service, and/or company affect the position presented in, or the review of, this manuscript.

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None.

DATA AVAILABILITY STATEMENT

Two datasets are employed in this study. The first one comes from [37], it is an underwater dataset synthesized by CycleGAN, and contains a total of 6128 image pairs with ground-truth (undistorted underwater images) and distorted underwater images. And each image has a corresponding reference image. From which we selected 4000 image pairs for training, and 100 image pairs from the remaining images for testing. The other one is the UIEB dataset [24], which contains 60 challenging real underwater images without corresponding reference images. To further verify the effectiveness of our mode, we grouped these images into five categories: greenish underwater images, bluish underwater images, yellow underwater images, shallow underwater images, and limited illumination underwater images.

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ORCID

Zhenghao Shi https://orcid.org/0000-0001-5291-9455
Yongli Wang https://orcid.org/0000-0001-9452-7513
Zhaorun Zhou https://orcid.org/0000-0001-8933-9761

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